

#HW 2

● Graded

Student

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 [View or edit group](#)

Total Points

84 / 100 pts

Question 1

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts There has been a mistake while you inter-converted the units. Your answer should be -4.63 Btu
- 0.5 pts Your final answer is not clear. You are supposed to write-4.63. The number 4 is not clear.
- 2 pts You were supposed to convert your answer to Btu
- 7 pts The answer has not been submitted.
- 2 pts Answer should be in Btu.

Question 2

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 7 pts Answer has not been submitted.
- 4 pts You have to find out the area of the plot. To calculate work you should multiply the volume difference with pressure. You have taken individual volumes and that should not be done.
- 2 pts You need to add up the areas. You have subtracted instead.
- 7 pts Answer has not been submitted.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts You have to calculate the work done in all the three steps and add them up. You have only done for the first step.
- 2 pts The work done will be negative in the first step. The total work should therefore be -2000 kJ.
- 2 pts It has been given $P(V^{1.5})$ constant. So $P = \text{constant}/(V^{1.5})$. You should apply that in integration of PdV in order to calculate work.
- 1 pt You should calculate the total work in all three steps.
- 2 pts There is an error in your calculation. The total work done will be -2000 kJ.
- 7 pts Answer has not been submitted.

Question 4

(no title)

14 / 15 pts

- 0 pts Correct

✓ - 1 pt In part b E2 will be 22 in order to get 15 as the difference.

- 4 pts ΔE is the difference of two energies. E1 is to be subtracted from E2. You should not add E1 and E2.
- 1 pt in part e, E1 should be +8 to get $\Delta E = 0$.
- 1 pt in c, ΔE should be -14 from $Q - W$, accordingly E2 should be -8 to get ΔE as -14.
- 7 pts Answer has not been submitted.

Question 5

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 5 pts Answer has not been submitted.
- 1 pt You should calculate the total heat transfer in the process.
- 2 pts There is a mistake while you changed units. Answer should be -50kJ.

Question 6

(no title)

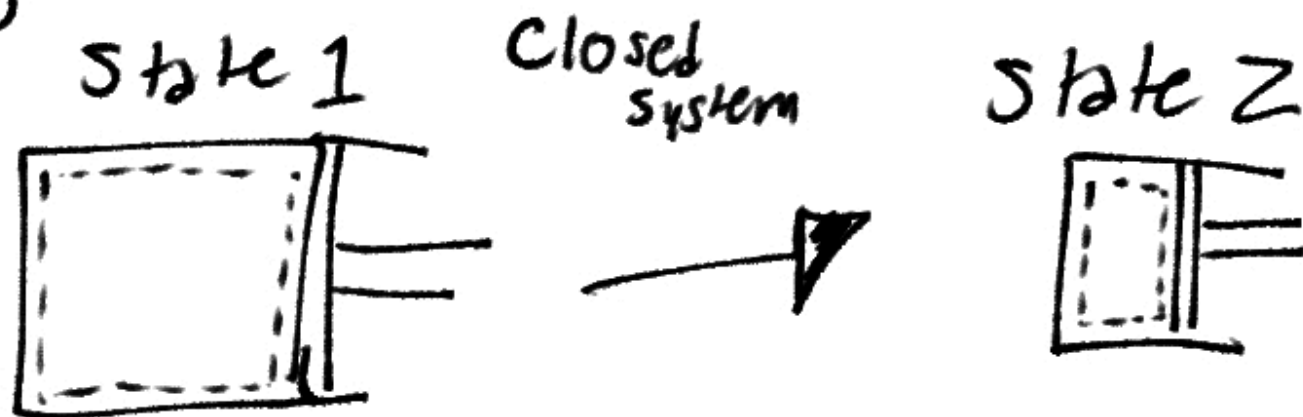
15 / 30 pts

- 0 pts Correct

✓ - 15 pts No essay submitted

Question assigned to the following page: [1](#)

2.13.18



$$P_1 = 5 \text{ lbf/in}^2$$

$$V_1 = 2.5 \text{ ft}^3$$

$$P_2 = 20 \text{ lbf/in}^2$$

$$V_2 = 0.5 \text{ ft}^3$$

$$P = 23.75 - 7.5V$$

work in BTU

$$W = \int_{V_1}^{V_2} P dV = \int_{2.5}^{0.5} (23.75 - 7.5V) dV = 23.75V - \frac{7.5V^2}{2} \Big|_{2.5}^{0.5}$$

$$= -25 \frac{\text{lbf} \cdot \text{ft}^3}{\text{in}^2} \times \frac{0.18505 \text{ BTU}}{1 \text{ lbf} \cdot \text{ft}^3 / \text{in}^2} = \boxed{-4.63 \text{ BTU}}$$

Question assigned to the following page: [2](#)

Z.13.22

$$\begin{aligned} \text{kJ} \cdot \text{m}^3 &= -((0.055 - 0.015)150 + (0.07 - 0.055)(100 + 50 \cdot \frac{1}{2})) \\ &= -7.875 \text{ kJ} \cdot \text{m}^3 = \boxed{-7.88 \text{ J}} \end{aligned}$$

Question assigned to the following page: [3](#)

2.13.24

Closed system

$$W = \int_{V_1}^{V_2} P dv = \int_1^4 \frac{10}{v^{1.5}} dv = 10 \text{ bar} \cdot \text{m}^3$$

$$10 \text{ bar} \cdot \text{m}^3 \times \frac{10^5 \text{ Pa}}{1 \text{ bar}} \times \frac{1 \text{ kJ}}{1000 \text{ J}}$$

$$10(1^{1.5}) = 10, \text{ Constant} = 10 \quad \frac{11}{1000 \text{ kJ}}$$

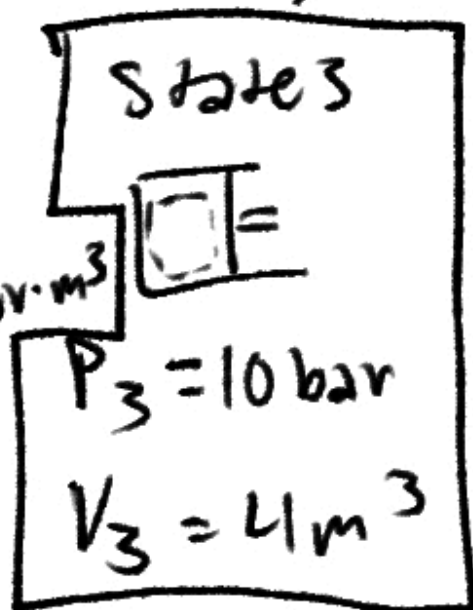
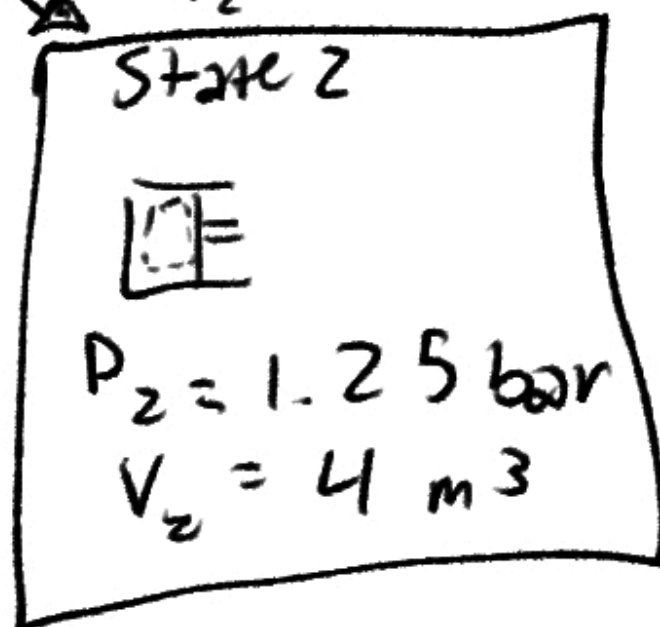
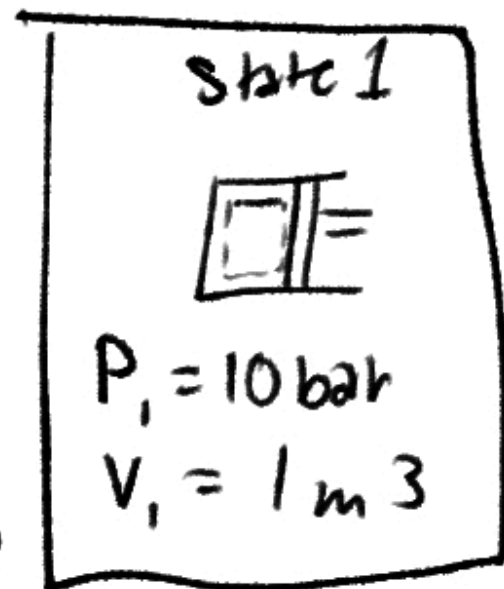
$$PV^{1.5} = 10$$

$$P = \frac{10}{V^{1.5}}$$

$$P = \frac{10}{4^{1.5}} = 4 \text{ bar}$$

$$PV^{1.5} = \text{Constant}$$

$$W_{1-2} = 1000 \text{ kJ}$$



$$W_{3-1} = \int_{V_3}^{V_1} P dv$$

$$= \int_4^1 10 dv$$

$$= 10v \Big|_4^1$$

$$= 10 - 40 = -30 \text{ bar} \cdot \text{m}^3$$

$$-30 \text{ bar} \cdot \text{m}^3 \times \frac{10^5 \text{ Pa}}{1 \text{ bar}}$$

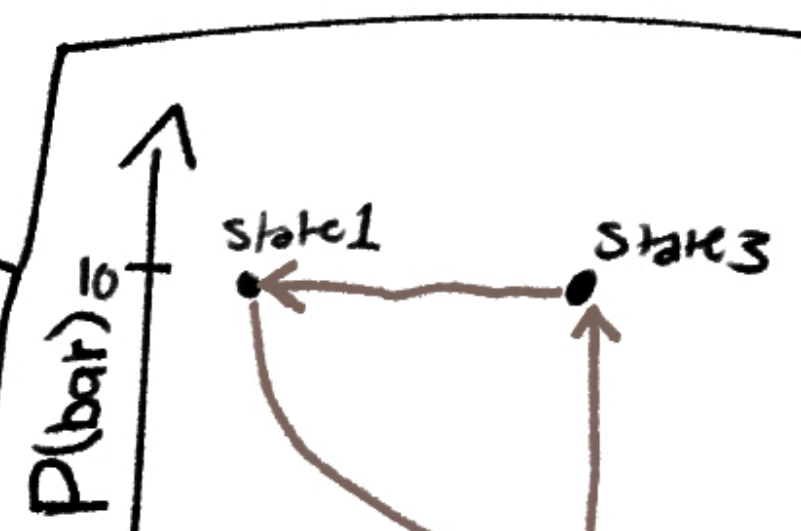
$$\times \frac{1 \text{ kJ}}{1000 \text{ J}} = -3000 \text{ kJ} = W_{3-1}$$

isobaric

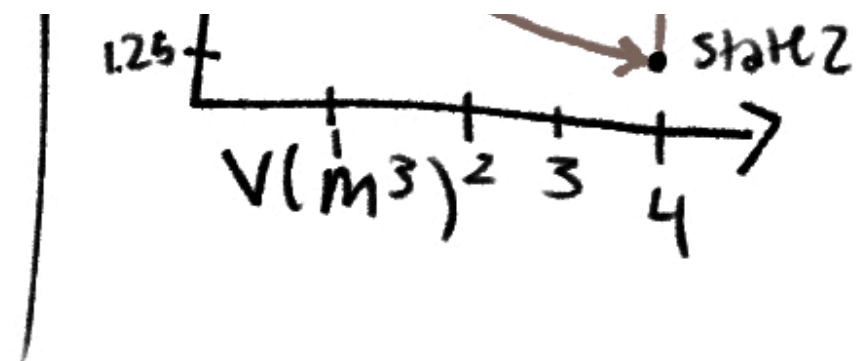
isochoric

$$W_{2-3} = 0, \text{ b/c isochoric process}$$

$$W_{\text{net}} = W_{1-2} + W_{2-3} + W_{3-1} = (1000 + 0 + (-3000)) \text{ kJ} = -2000 \text{ kJ} = W_{\text{net}}$$



Question assigned to the following page: [4](#)



2.13.43

Process	Q	W	E_1	E_2	ΔE
a	+40	-25	+15	+30	+15
b	+27	+12	+7	19	+15
c	-4	+10	+6	-8	-14
d	+16	-10	-10	+10	+20
e	+3	+3	+8	+8	0

$$\Delta E = Q - W$$

4 2 1 3

Question assigned to the following page: [5](#)

2.13.44 - Closed system

$$\underline{W} = 0.147 \text{ kJ/kg} \cdot 10 \text{ kg} = \underline{1.47 \text{ kJ}}$$

$$V_1 = 15 \text{ m/s} \quad V_2 = 30 \text{ m/s}$$

$$h_1 = 0 \text{ m} \quad h_2 = 50 \text{ m}$$

$$\underline{\Delta U} = -5 \text{ kJ/kg} \cdot 10 \text{ kg} = \underline{50 \text{ kJ}}, \quad g = 9.7 \text{ m/s}^2$$

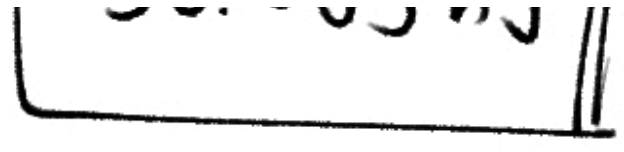
$$\Delta E = Q - W \rightarrow \underline{\Delta E + W = Q}$$

$$\underline{\Delta KE} = \frac{1}{2} m V_2^2 - \frac{1}{2} m V_1^2 = 3375 \text{ J} = \underline{3.375 \text{ kJ}}$$

$$\underline{\Delta PE} = m g (h_2 - h_1) = -4850 \text{ J} = \underline{-4.850 \text{ kJ}}$$

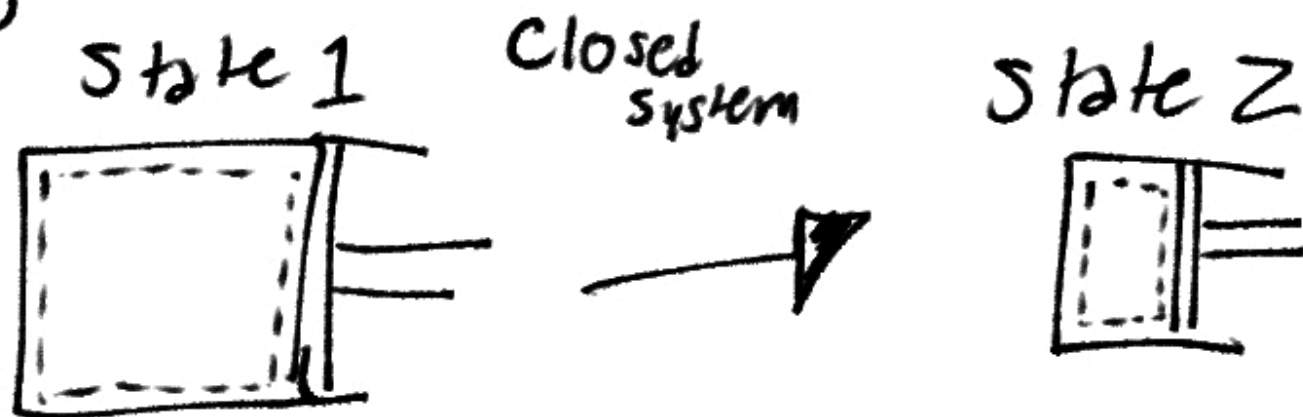
$$\underline{\Delta E} = \Delta U + \Delta KE + \Delta PE = 3.375 + (-4.850) + (-50) = \underline{-51.475 \text{ kJ}}$$

$$\rightarrow Q = (-51.475 + 1.47) \text{ kJ} = \underline{\underline{-50.005 \text{ kJ}}}$$



Question assigned to the following page: [6](#)

2.13.18



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